Feature selection

In this lesson we will need synthetic data with ~8 variables to reproduce ISLP figures 6.4 (left panel) and 6.6 (left panel).

Set up: We have p predictor variables $X_1, \dots, X_p$ and we want to model a response variable Y

[This is a recap of what fitting linear model with least squares means]

Basic option: Include all p predictors $X_1, \dots, X_p$ in your model

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

Fit your model using least squares to obtain estimates $\hat{\beta}_0, \dots, \hat{\beta}_p$ of the coefficients:

$$Y = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_p X_p.$$

Issue: Including all variables is not guaranteed to give you the best model. How do we find a "good enough" subset of variables to fit the model?

[This is a recap about best subset selection and a reminder that R^2 increases with the number of predictors so we need to use other metrics to compare]

[This is going over forward/backward stepwise selection to select a subset of variables that could do a better job than the full set of variables. Cover: AIC, adjusted R^2.]

Another option: From the predictors $X_1, \dots, X_p$, find a subset of q predictors $X_{i_1}, \dots, X_{i_q}$ that maximizes a metric.

Fit the final model on this subset of predictors using least squares.

$$Y = \hat{\beta}_0 + \hat{\beta}_1 X_{i_1} + \dots + \hat{\beta}_q X_{i_q}.$$

What if instead of explicitly looking for a subset of predictors, we could alter the fitting method to find that subset of good predictors for us?

Check-in: In the linear model $Y = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_p X_p$, how would you "remove" the variable X_1 by changing the coefficient estimates?

Idea: fit a model with all predictors using a method that shrinks the coefficient estimates towards 0.

Shrinking = regularization = constraining

Instead of directly minimizing the residual sum of squares

$$\text{minimize}(RSS)$$

regularization introduces a penalty term that takes into account the number of variables.

$$\text{minimize}(RSS + \text{penalty}).$$

The penalty parameter constrains the size of the coefficients so they can only increase if accompanied by a decrease in SSE.

We will see two methods to introduce penalty: Ridge and lasso.

Ridge regression

In Ridge regression the coefficients are estimated by minimizing

$$RSS + \underbrace{\lambda \sum_{i=1}^{p} \beta_i^2}_{\text{penalty}},$$

where $\lambda \geq 0$ is a tuning parameter that is determined separately.

The shrinking penalty is small when $\beta_1, \dots, \beta_p$ are close to zero, so it has the effect of shrinking the estimates of β_j towards zero.

The tuning parameter λ controls the relative impact of the RSS and the penalty on the regression coefficient estimates.

Ridge regression coefficients can change substantially when multiplying a predictor by a constant. Best to apply Ridge regression after standardizing all the predictors using the formula

$$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2}},$$

so they are all on the same scale.

Q: Do we also need to standardize the predictors for lasso?

[Here we need a plot of how the standardized coefficients change as $\lambda \to \infty$ like ISLP figure 6.4, the left panel]

- When $\lambda = 0$, the penalty has no effect and the coefficient estimates will be the same as the least squares.
- As $\lambda \to \infty$, the effect of the penalty grows and the coefficients will approach zero

Check-in: make the last two bullet points into a check in.

- Each value of λ will produce a different set of coefficients

Check-in: ask them to describe how they would use CV to find a good λ.

Q: Are the ridge regression coefficients interpretable somehow?

*In another iteration I want to have more discussion about variance / bias as λ grows

Selecting the tuning parameter

Here we have a short section on using cross-validation to select λ.

The Lasso

In the lasso regression, the coefficients are estimated by minimizing

$$RSS + \underbrace{\lambda \sum_{j=1}^{p} |\beta_j|}_{\text{penalty}}.$$

The lasso penalty has the effect of forcing some of the coefficient estimates to be exactly equal to zero when the tuning parameter λ is sufficiently large. So the lasso performs variable selection.

Check-in:

1. When $\lambda = 0$, how are the coefficient estimates obtained with the lasso and the least squares fit related?
2. What is the value of the coefficients when λ becomes very, very large?

Comparing lasso and ridge

Ridge regression will include all p predictors in the final model. The ridge penalty $\lambda \sum \beta_i^2$ will shrink all coefficients towards zero, but won't set any equal to zero. If p is very large, then it can be hard to make sense of that many variables.

Neither lasso nor ridge regression universally dominate the other.

- We might expect lasso to perform better when a relatively small number of predictors have substantial coefficients, and the remaining predictors have coefficients that are very small or equal zero.
- Ridge will perform better when the response is a function of many predictors, all with coefficients of roughly equal size.
- But the number of predictors associated to the response is never known a priori for real data sets!
- Cross-validation can be used to compare both approaches.